Biochemistry Cume 2018

Catalysis of the cleavage of peptide bonds in small peptides by a proteolytic enzyme at pH=7 is described in the following table. Use the data to answer questions 1 and 2 below. The arrow, ↓, indicates the location of peptide cleavage.

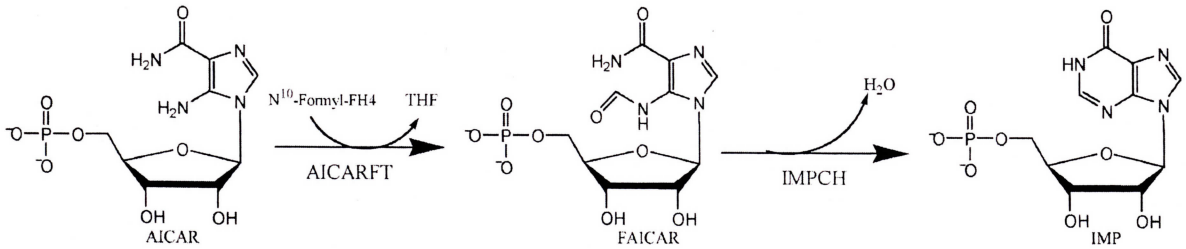
Substrate	K_M(mM)	k_{cat} (s^{-1})
Gln-Ser-Thr-Lys-↓Gly	80	8000
Gln-Ser-Thr-Arg-↓Val	400	10000
Gln-Ser-Thr-Gln-↓Gly	0.0003	0.003
Gln-Ser-Thr-Arg-↓Ala	0.1	8
Gln-Ser-Thr-Lys-↓Ile	60	360

1. [25pts] If a mixture of these peptides were presented to the enzyme with the concentration of each peptide being the same, which peptide would be cleaved most rapidly? Provide your reasoning for full credit. Providing no reasons for answer will result in 0 pt.

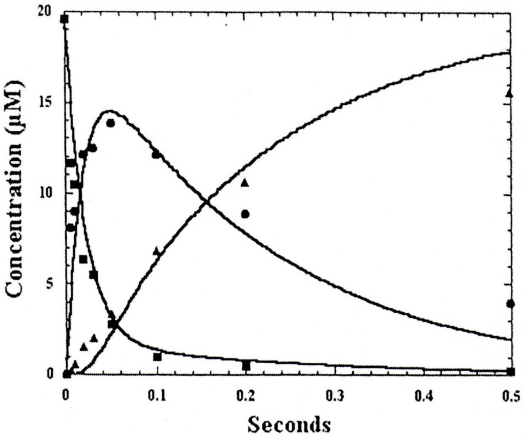
2. [25pts] On the basis of these data, estimate the features of the amino acid sequence that dictate the specificity of enzyme. Then identify the peptide that would be cleaved most rapidly by the same enzyme. Provide your reasoning for full credit. Providing no reasons for answer will result in 0 pt.

   Gln-Ser-Thr-Arg-Gly
   Gln-Ser-Thr-Gln-Ala
   Gln-Ser-Thr-Arg-Ile
   Gln-Ser-Thr-Arg-Leu
   Gln-Ser-Thr-Asp-Gly

3. [25pts] Human 5-Amino-4-imidazolecarboxamide ribonucleotide transformylase/IMP cyclohydrolase (hATIC) is a bifunctional protein possessing two enzymatic activities that sequentially catalyze the last two steps in the pathway for *de novo* synthesis of inosine 5′-monophosphate. This bifunctional enzyme is of particular interest because of its potential as a chemotherapeutic target.



Single turnover time course for the bifunctional hATIC reaction is plotted. The hATIC enzyme (80 μM) was mixed with [^{3}H]AICAR (20 μM) and N^{10}-formyl-FH_4 (500 μM). The disappearance of [^{3}H]AICAR (▪) and the appearance of [^{3}H]FAICAR (●) and [^{3}H]IMP (□) were monitored. The authors concluded that hATIC does not channel its substrate. (JBC, 2002, 22168)



If the enzyme were channeling the substrate, what would the time course plot look like. Provide your reasoning for full credit. Providing no reasons for answer will result in 0 pt.

4. [25pts] Gout resulting from the *de novo* overproduction of purines (defect A) can potentially be distinguished from gout caused by impaired excretion of uric acid (defect B) by feeding a patient ^{15}N-labeled glycine and determining the distribution of ^{15}N in his or her excreted uric acid. What isotopic distributions are expected for each type of defect? Provide your reasoning for full credit. Providing no reasons for answer will result in 0 pt.